VOCES DE LA SIERRA

Bridging Languages, Preserving Cultures

An Offline Bidirectional Translator & Cultural Teacher

Español ↔ Náhuatl

Powered by Gemma 4 E2B · Fine-tuned with Unsloth · Edge AI with llama.cpp

Gemma 4 Good Hackathon 2026

Kaggle × Google DeepMind

Tracks: Digital Equity & Inclusivity · Future of Education

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The Problem: When Your Language Makes You Invisible

In the highland markets of Mexico's Sierra Norte, a scene repeats itself every day that most of the world never sees. A woman walks down from her village carrying herbs, textiles, and knowledge passed through generations. She speaks Náhuatl — a language older than the conquest, older than the borders drawn on modern maps. But when she reaches the market stall, when she needs to negotiate a price, explain a symptom to a doctor, or understand a government form — she becomes invisible.

Not because she lacks intelligence. Not because she lacks something to say. But because the world around her operates exclusively in a language she was never given the tools to fully master.

This is not a hypothetical scenario. This is the lived reality of about 1.7 million Nahuatl speakers in Mexico, many of whom live in communities without reliable internet access, without smartphones capable of running cloud-based translation services, and without the economic resources to access professional interpretation.

Major commercial translation platforms do not currently provide robust support for Náhuatl. These 1.7 million people exist in a technological blind spot — too small a market for commercial interest, too linguistically complex for simple rule-based systems, and too disconnected from the internet for cloud-dependent solutions.

Voces de la Sierra was built to change that.

The Solution: AI That Works Where It's Needed Most

Voces de la Sierra is a bidirectional translator and cultural teaching assistant that operates 100% offline on consumer hardware. It translates between Spanish and Náhuatl in both directions, and when a user wants to learn rather than just translate, it switches into a patient, culturally-aware teaching mode that explains vocabulary, grammar, and the meaning behind the words.

The system has two roles. As a Translator, it provides direct, precise translations between both languages. As a Maestro Docente (Teaching Master), it explains how words are used, their cultural significance, and helps users practice pronunciation — bridging not just languages but worldviews.

Technical Architecture

Model Selection: Gemma 4 E2B

We chose Gemma 4 E2B-it (Edge 2 Billion parameters) for three critical reasons. First, it is designed for edge deployment — the E in E2B stands for Edge, meaning it was architecturally optimized to run on resource-constrained devices. Second, its multilingual pretraining provides a useful starting point for adapting the model to low-resource languages like Náhuatl. Third, its 4-bit quantized variant fits within 1.5GB of RAM, making it viable for low-end Android devices.

Fine-Tuning Pipeline

Training Infrastructure: Google Colab T4 GPU (14.5GB VRAM) with Unsloth optimization framework.

Technique: QLoRA (Quantized Low-Rank Adaptation) with r=8, alpha=8, targeting all attention and MLP projection layers (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj). This trains only 0.29% of the model's 5.1B parameters (14.9M trainable parameters), making the entire training possible on free-tier hardware.

Dataset: 20,028 Spanish-Náhuatl parallel pairs from the Axolotl corpus (SomosNLP Hackathon 2022), which includes conversational phrases and biblical text from the Bible-uedin parallel corpus. After quality filtering using morphological validators (detecting Náhuatl-specific morphemes like -tl, -tzin, -tli), 16,822 clean pairs remained.

Data Augmentation Strategy: Each clean pair generates 4 training conversations: (1) Spanish→Náhuatl translation, (2) Náhuatl→Spanish translation, (3) Docent explanation Esp→Náh with cultural context, (4) Docent explanation Náh→Esp with usage examples. This produces 67,288 training examples from the original 20,028 pairs.

Training Configuration: 10,000 steps with batch size 1, gradient accumulation 4 (effective batch 4), cosine learning rate schedule starting at 2e-4, AdamW 8-bit optimizer. The training loss went from 1.14 to a minimum of 0.0368 (step 8,574), with a final sustained average of ~0.20 in the last 1,000 steps, with the model learning to generate authentic Náhuatl morphology including correct use of absolutive suffixes, honorific particles, and verbal conjugation patterns.

Deployment Architecture: 100% Offline

The trained adapter is exported to GGUF format using llama.cpp and deployed locally via llama-server, providing a web-based chat interface accessible at localhost:8080 — no internet connection required. The complete system consists of: the Gemma 4 E2B base model in Q4_K_M quantization (~1.8GB), the LoRA adapter in GGUF format (~23MB), and llama-server as the inference engine. Total storage requirement: under 2GB.

For the hackathon demo, inference is also demonstrated directly in Google Colab, where the model successfully produces Náhuatl translations including real morphological patterns. The model correctly translates "el agua es vida" to Náhuatl using the authentic word "atl" (water) and generates culturally relevant vocabulary.

Training Resilience System

A critical engineering challenge was training on Google Colab's free tier, which limits GPU sessions to 5 hours. We designed a checkpoint persistence system that saves model state to Google Drive every 500 training steps. Each new session automatically detects the last checkpoint and resumes training from that point. This allowed us to accumulate 10,000 training steps across multiple sessions without losing progress — effectively turning a 5-hour limitation into an unlimited training pipeline.

Impact & Scalability

Who This Serves

The primary users are Náhuatl-speaking communities in Mexico's Sierra regions who need to communicate with Spanish-speaking institutions: hospitals, government offices, schools, and markets. Secondary users include educators working in bilingual indigenous schools, linguists and cultural preservation organizations, and anyone interested in learning Náhuatl as a second language.

Why Offline Matters

In the Sierra Norte of Puebla, the Sierra de Oaxaca, and the Huasteca region, internet connectivity ranges from unreliable to nonexistent. A cloud-based translator is useless where there are no clouds to connect to. Voces de la Sierra runs entirely on the device — no API calls, no server dependencies, no data leaving the user's hands. This also addresses a critical privacy concern: indigenous communities have historically had their linguistic data extracted without consent. Our system keeps all interactions local.

Scalability to Other Languages

The architecture is designed to be adaptable to other low-resource language pairs, subject to the availability of aligned data, community validation, and linguistic adaptation. Mexico alone has 68 indigenous language groups with 364 variants. Globally, UNESCO states that at least 40% of the world's more than 7,000 languages are threatened with extinction. Each one represents a unique way of understanding the world. Voces de la Sierra provides a template for preserving them through accessible AI.

Challenges & Learnings

Gemma 4 is a cutting-edge model released weeks before this hackathon, and the tooling ecosystem is still maturing. We encountered and solved several significant challenges during development:

- **VRAM constraints:** Gemma 4 E2B's multimodal architecture (including audio and vision towers) consumes nearly all of the T4's 14.5GB VRAM. We solved this by using Unsloth's official T4-optimized configuration with max_seq_length=1024 and lora_dropout=0 for full fast-patching.
- **GGUF export pipeline:** The standard save_pretrained_gguf fails for Gemma 4 due to bitsandbytes quantization incompatibility with llama.cpp. We developed a custom export pipeline that extracts text-only LoRA weights, renames tensor keys to match llama.cpp conventions, and converts via convert_lora_to_gguf.py with a 16-bit base model reference.
- **Náhuatl dialectal variation:** The Axolotl corpus contains multiple dialectal variants (Classical, Huasteco, Sierra) without labels. We implemented a morphological validator using regex patterns for diagnostic Náhuatl morphemes (-tl, -tzin, -tli) to filter entries where the Náhuatl column contained Spanish text.
- **Chat template alignment:** Gemma 4 uses <start_of_turn>/<end_of_turn> tokens with native system prompt support. Misalignment between training and inference templates produces gibberish output. We built the training templates manually using verified token strings rather than relying on apply_chat_template which was unreliable for the multimodal tokenizer.

Future Roadmap

1. **Voice Input:** Gemma 4 E2B natively supports audio input through its USM encoder. When Ollama/llama.cpp add audio support, users will be able to speak in Náhuatl and receive translations — critical for a predominantly oral language.
2. **Mobile deployment:** Package as a standalone Android APK using llama.cpp's Android bindings, enabling direct installation on low-end phones without Docker or server configuration.
3. **Expanded datasets:** Partner with indigenous language organizations (INALI, CIESAS) to obtain larger, dialect-labeled corpora for more accurate regional translations.
4. **Multi-language expansion:** Apply the same pipeline to Mixtec, Zapotec, Maya, and other endangered Mexican indigenous languages using the same Gemma 4 base.

Conclusion

A language is not just a way of speaking. It is a way of thinking, of understanding the relationship between water and life, between earth and identity, between ancestors and the present moment. When a language disappears, an entire philosophy of existence goes with it.

Voces de la Sierra is not just a translator. It is a statement that the most powerful AI in the world should work for the people who have the least. That edge computing and open models like Gemma 4 can reach the places where cloud services never will. And that a woman walking down from the Sierra to sell her herbs in the market deserves the same technological dignity as anyone with a broadband connection.

The code is open. The model weights are published. The architecture is replicable. And the next voice preserved could be the one that was about to disappear forever.

Technical Specifications Summary

Base Model: Google Gemma 4 E2B-it (5.1B parameters, 4-bit quantized)

Fine-tuning: QLoRA via Unsloth | r=8, alpha=8, dropout=0 | 14.9M trainable params (0.29%)

Dataset: 67,288 training examples from 20,028 Esp↔Náh pairs (Axolotl + Bible UEDIN)

Training: 10,000 steps | Colab T4 free tier | Checkpoint persistence via Google Drive

Min Loss: 0.0368 (step 8,574) · **Final avg:** ~0.20 (last 1,000 steps) · **Starting:** 1.14

Deployment: llama.cpp server | 100% offline | ~2GB total storage | Web UI at localhost:8080

License: Apache 2.0 (compatible with Gemma 4 license)

Repository: https://github.com/AnthonyJTR22/voces-de-la-sierra.git

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